DEEPAK KATUKURI

DATA AUTOMATION · PYTHON & SQL · AI AGENTS



CONTACT

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TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, Core Java, C++

Data & Automation: Pandas, NumPy, data-migration scripts, data validation, backfill scripts, pytest, Matplotlib, Seaborn, Recharts dashboards, Jupyter

Databases: PostgreSQL, MySQL, pgvector, Chroma, SQLAlchemy, Alembic

AI Agents & LLM: LangGraph, LangChain, Claude API, agentic workflows, prompt design, Retrieval-Augmented Generation (RAG), hybrid retrieval (BM25 + dense), cross-encoder reranking, Ollama, Claude Code

ML: Scikit-Learn, PyTorch, TensorFlow, Hugging Face, classification, ensembles

Full-Stack & Cloud: FastAPI, Flask, React, Docker, AWS (EC2, RDS), Kubernetes (K3s), GitHub Actions, CI/CD, Vercel, Render

Tools: Git/GitHub, Linux (Ubuntu)

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech in Computer Science & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL (Database Management)
Core Java (Software Development)
OneShield (P&C Insurance)

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

Data Science MSc candidate and Software Engineer with 2.5 years of production data work: migrating, validating and consolidating policy, claims and premium records for insurance clients in PL/SQL. Automates recurring data tasks in Python and SQL, and builds LLM agents that check their output against a verified source: a 16-node LangGraph pipeline and a RAG retriever validated on a 55-question golden set. Ships dashboards and tested code (99-test pytest suite). Looking for an English-speaking working student role in data, automation and AI in Germany.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024

ValueMomentum · clients: Hiscox, Philadelphia Insurance

- Consolidated policy, claims, premium and customer history for Philadelphia Insurance's Surety Bonds line from a legacy policy admin system into OneShield PAS by writing and validating the PL/SQL migration scripts across customer, policy (PID), claims and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty (the OneShield object copy failed silently when another bond shared that bond type under a different bond ID). Automated a weekly backfill script that kept records current while a permanent fix was pending.
- Resolved 100+ production defects across two carrier platforms, Hiscox (P&C insurance) and Philadelphia Insurance (surety bonds). Turned underwriting and claims requirements into technical specs, and ran QA/UAT test cases and CI/CD deployments across the full policy lifecycle.

ML Intern
Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it on IBM Watson ML. Presented the results to non-technical reviewers with Matplotlib and Seaborn charts.

PROJECTS

JobPilot: Multi-Agent Document Automation Pipeline

Python · LangGraph · Claude API · Chroma · pytest · github.com/Deepak589/Job_hunt_Agentic_AI

- Automated a manual document-tailoring workflow with a 16-node LangGraph agent pipeline that diagnoses an input document, rewrites it, and blocks any claim without evidence in a Chroma-backed profile store — checkpointed for crash recovery, cost-tracked per run, and held honest by 238 pytest tests across 36 files.

RA Nutrition & Lifestyle App: Data Logging to Weekly Dashboard

React · FastAPI · PostgreSQL · Recharts · Docker · live at ra-project-indol.vercel.app · github.com/Deepak589/RA_project

- Shipped a full-stack health app — React/Vite frontend on Vercel, FastAPI + PostgreSQL backend on Render — that turns daily meal and symptom logs into a rule-based recommendation and a weekly analytics dashboard, backed by a 99-test pytest suite.

RAG Pipeline: Hand-Rolled Retrieval to Corrective RAG (CRAG)

Python · LangChain · pgvector · BGE-M3 · Corrective RAG · github.com/Deepak589/RAG_pipeline

- Built retrieval from scratch through 6 gated stages — hybrid BM25+dense, then framework-validated on LangChain (caught a silent 0.06-recall regression from a default tokenizer) — and added a corrective RAG loop that re-retrieves on a bad match and refuses when the corpus can't answer, reaching 100% correct refusal and 69.5% end-to-end correctness.

CloudNotes: Kubernetes & AWS Deployment Platform

AWS EC2 + RDS · K3s · GitHub Actions · Traefik · MSc cloud computing, team of 3 · github.com/Deepak589/cloudnotes

- Replaced a team's manual deploys with a GitHub Actions pipeline to AWS Kubernetes (K3s) — build, push, SSH, zero-downtime rollout — autoscaling 2 to 5 pods at 60% CPU; owned the Kubernetes, CI/CD and AWS layers on a 3-person team.